Introduction about some work

Zhongqi Xiu

2024.7.26

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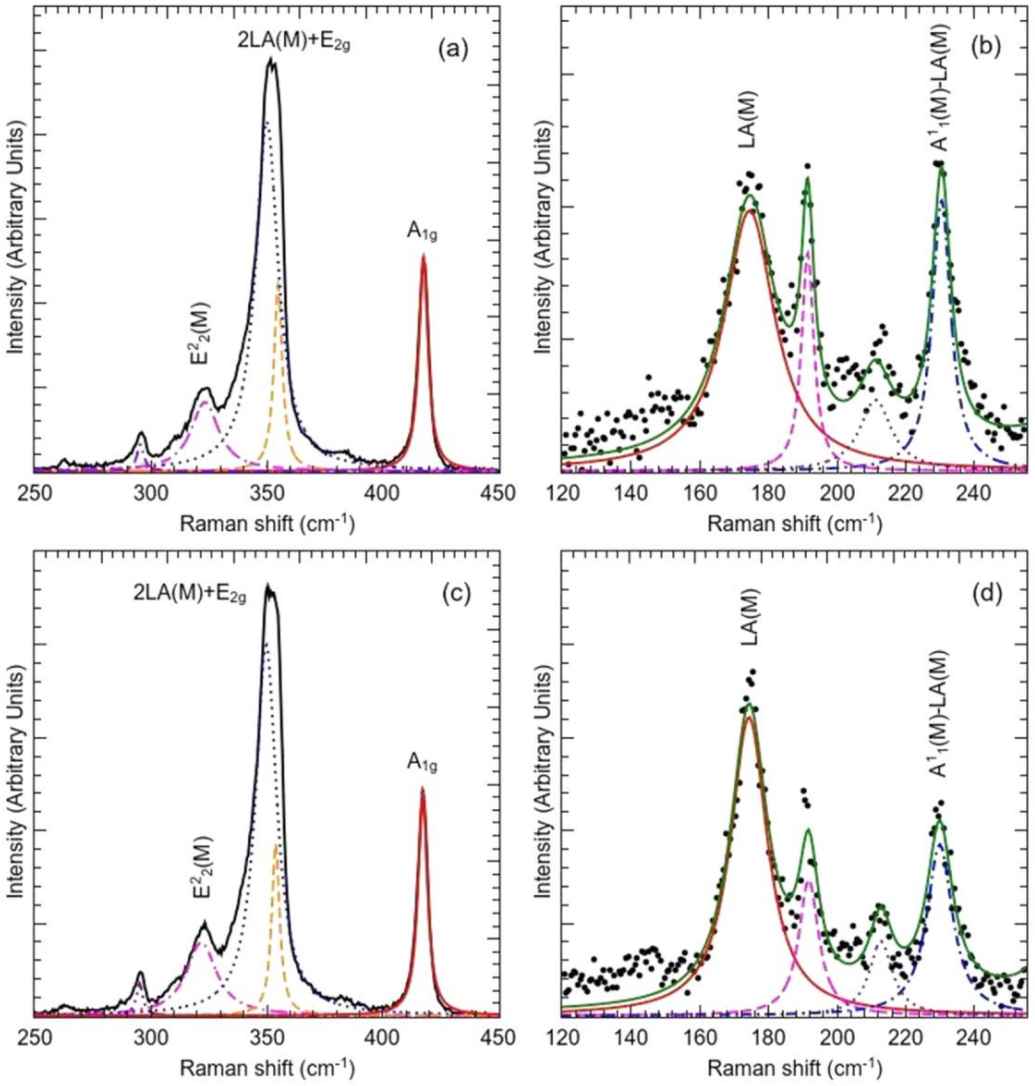
Luminescence enhancement and Raman characterization of defects in WS_2 monolayers treated with low-power N_2 plasma

André do Nascimento Barbosa[a,b,*], Cesar Augusto Diaz Mendoza[a], Neileth Johanna Stand Figueroa[a], Mauricio Terrones[b,c,d,e,f,g], Fernando Lázaro Freire Júnior[a]

Sample preparation : CVD

Plasma treatment : bombard, generate by RF coil, vacuum

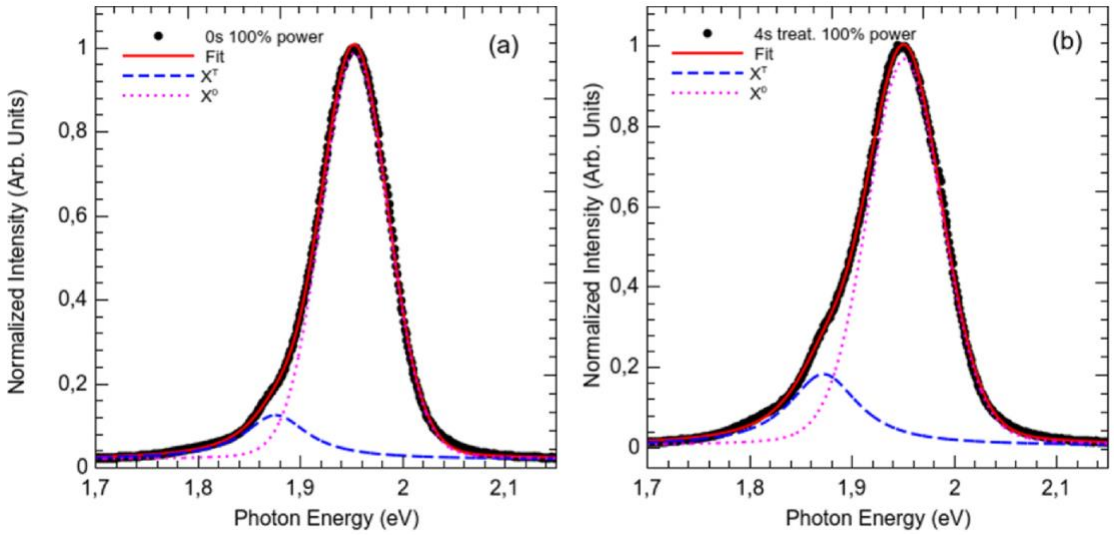
Raman(514.5nm, 488nm): LA(M) --- density of defects
$E_{2g}A_{1g}$ redshift --- nitrogen in sample, lattice constants



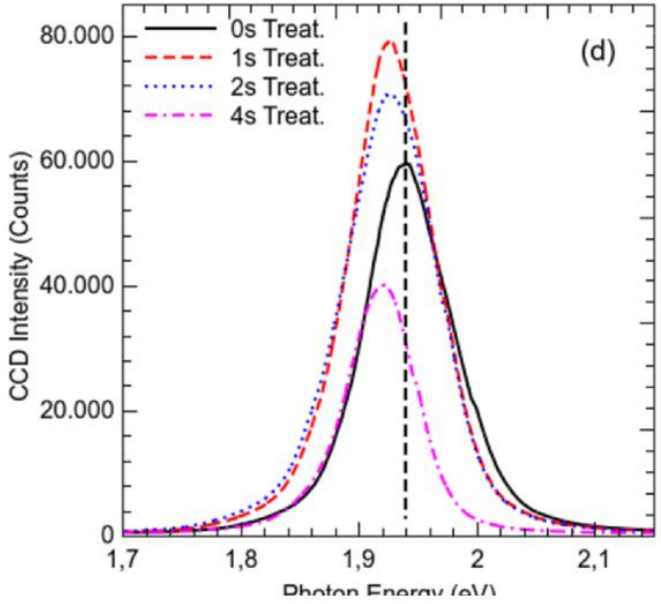
Sample	LA(M)			2LA(M)			E_{2g}
	Position (cm^{-1})	FWHM (cm^{-1})	Area	Position (cm^{-1})	FWHM (cm^{-1})	Area	Position (cm^{-1})
0 s	174,8	7,5	89	350.5	4.7	1148	355.3
1 s	174.8	6.8	97	350.3	4.7	1031	355.0
2 s	174.2	5.9	109	345.0	4.4	1015	354.7
4 s	174.7	5.4	110	349.6	4.3	935	354.5

https://doi.org/10.1016/j.apsusc.2020.147685

PL(473nm, 1µm spot, 2-0.02mW)



Normalized by the intensity of the first-order Raman Si peak



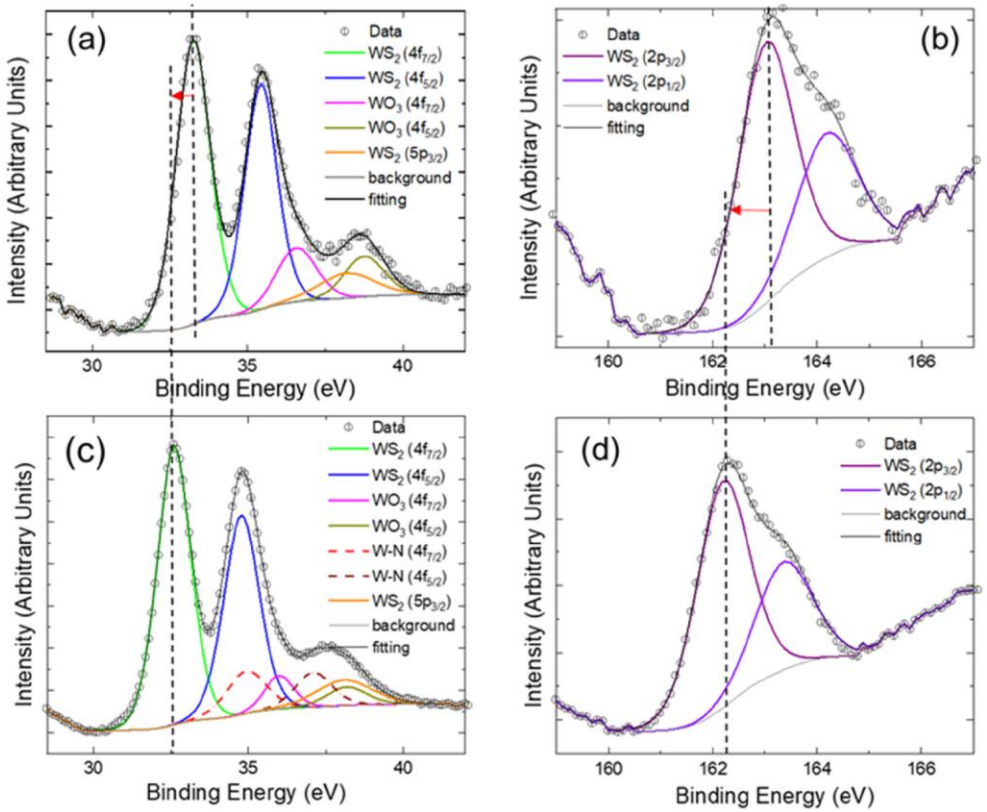
PL from recombination of electron–hole pairs is not strongly red-shifted---band structure not radically affected

Micro redshift originates due to the inclusion of discrete defect states inside the bandgap.

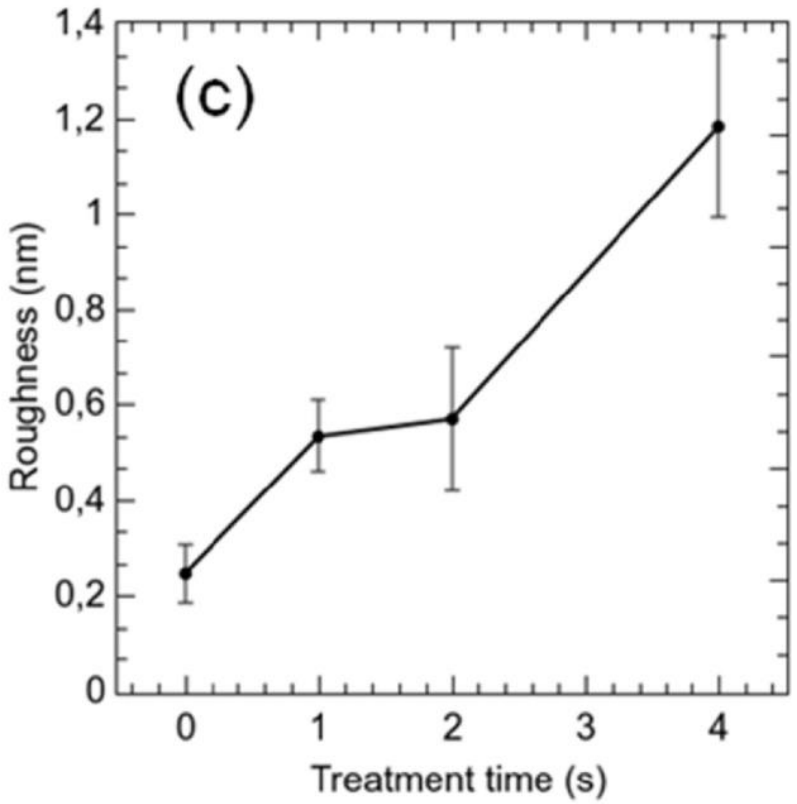
PL parameters related to the measured samples with 100% laser power.

Treatment Time (s)	FWHM (10^{-2} eV)	CCD mean Intensity (10^3 Counts)	Peak Energy (eV)	X^0/X^T Ratio
0	6.65	59	1.94	8.16
1	7.52	79	1.92	7.13
2	7.63	72	1.92	6.82
4	7.22	32	1.91	5.33

XPS spectra related to core levels of the W4f and S2p



The redshift observed for both S and W peaks after the plasma exposition can be attributed to the substitution of S by N.



The lower luminescence may be related to increase in roughness since the treatment is degrading the monolayers

2

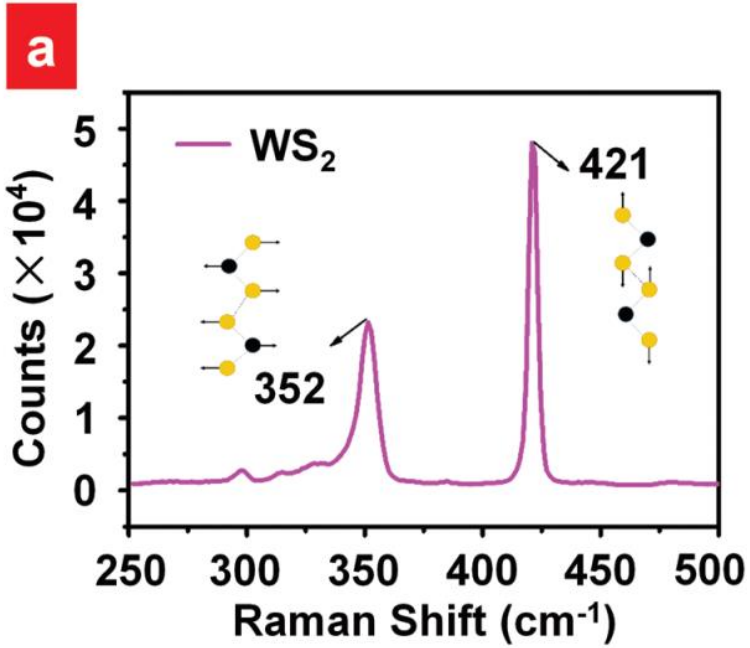
A Facile and Effective Method for Patching Sulfur Vacancies of WS_2 via Nitrogen Plasma Treatment

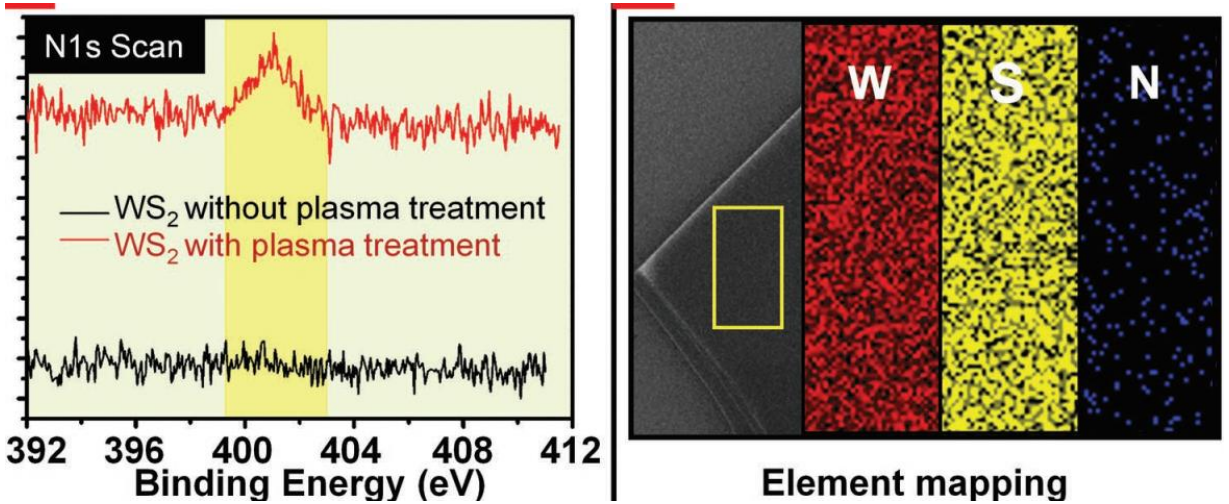
Jianfeng Jiang, Qinghua Zhang, Aizhu Wang, Yu Zhang, Fanqi Meng, Congcong Zhang, Xianjin Feng, Yuanping Feng, Lin Gu, Hong Liu,* and Lin Han**

Sample preparation : mechanical exfoliation

Plasma treatment : PDC-32G plasma cleaner under a high RF LEVEL for various durations. A mechanical pump was used to evacuate for 10 min; then, nitrogen gas kept flowing for 10 min to stabilize the internal pressure. The power was turned on to record the glow time.

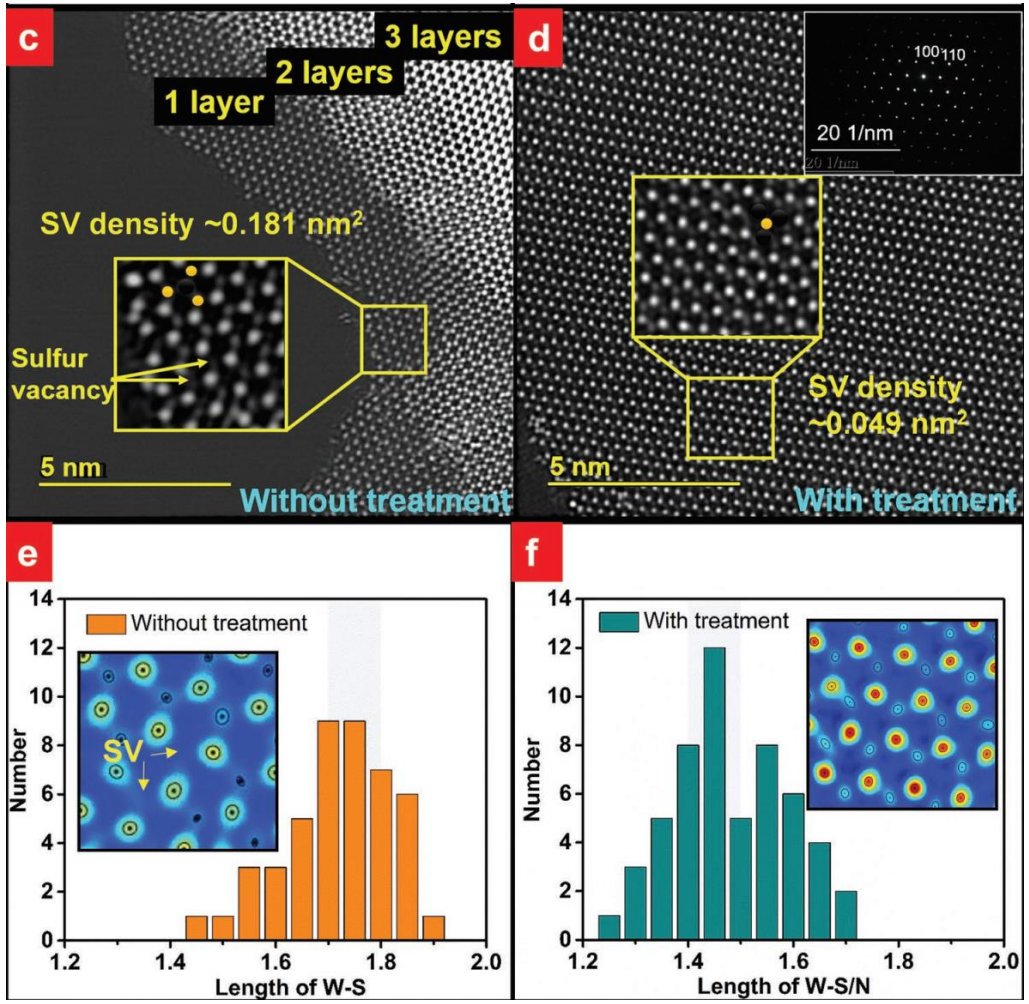
Raman(right): 532nm
XPS(down): N1s peak 401.1eV(pyrrolic-N)





https://doi.org/10.1002/smll.201901791

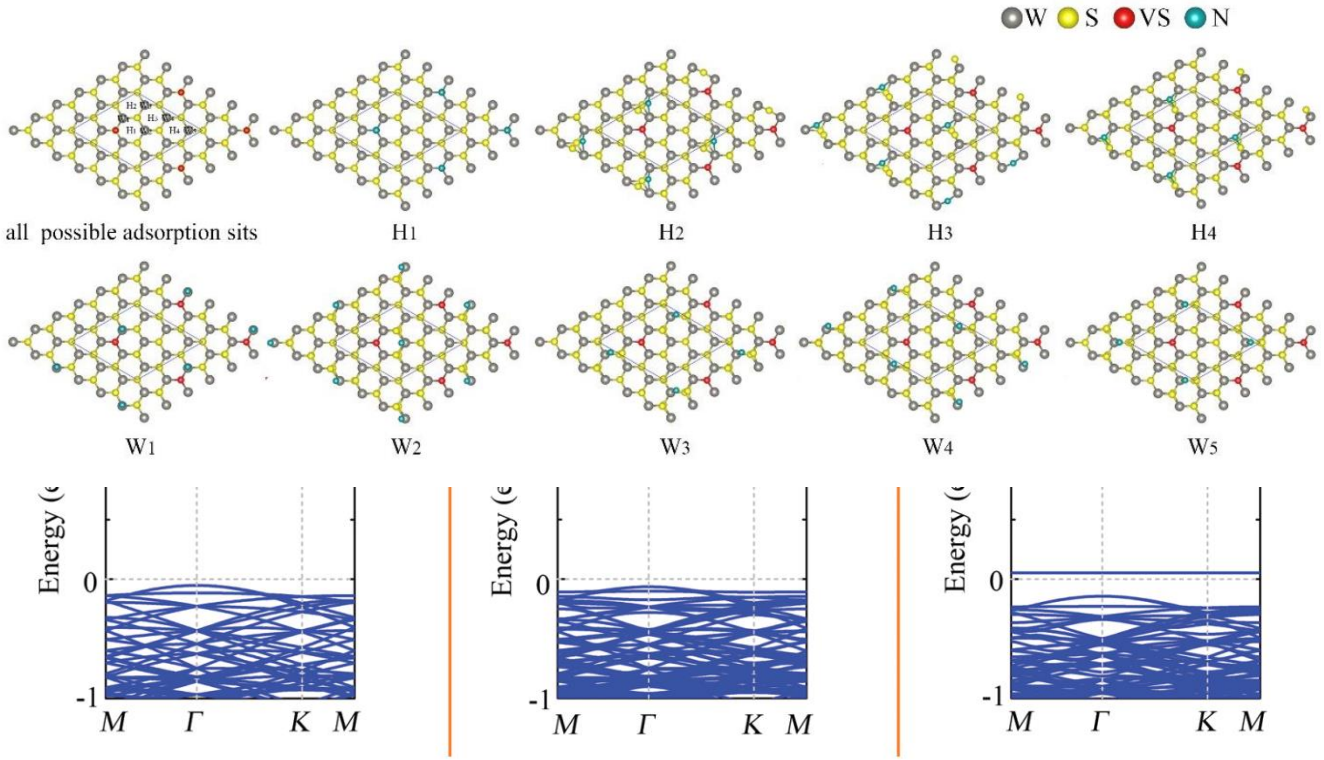
Atomic structures via STEM

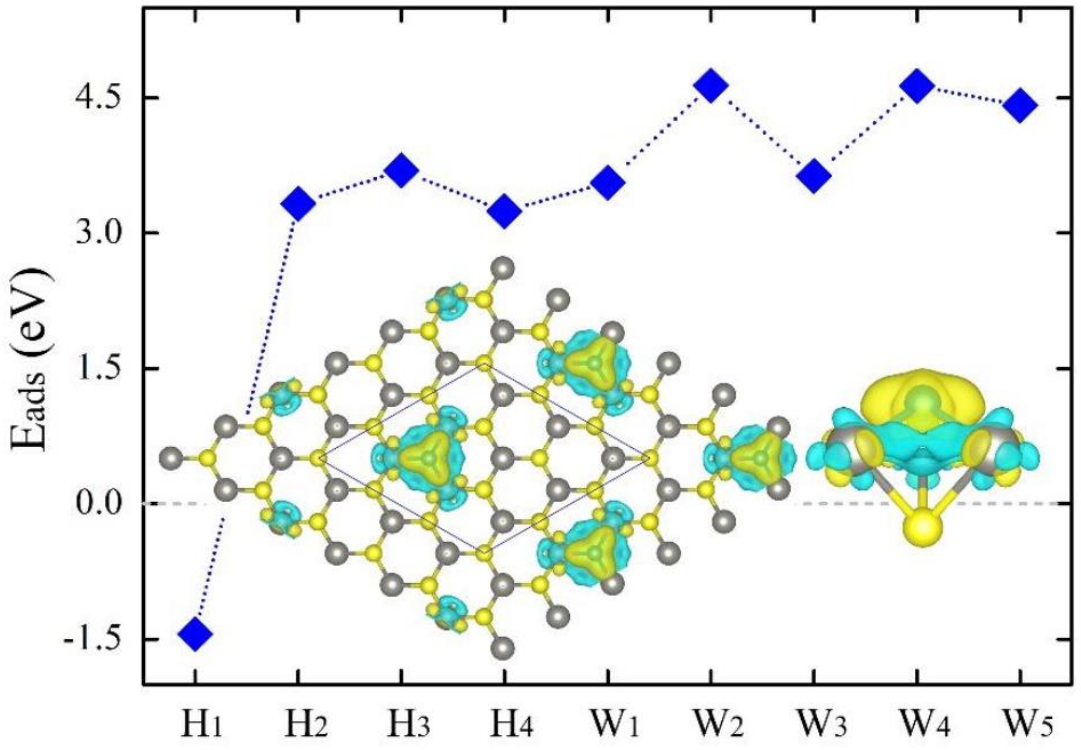


After treatment, the vacancy density is lower and the length of W-S/N shorten.

SVs were repaired and adsorbed by N atoms by forming W-N bonds and N doping can reduce the work function of WS2 theoretically.

First-principles calculations





Adsorption energies of the sites

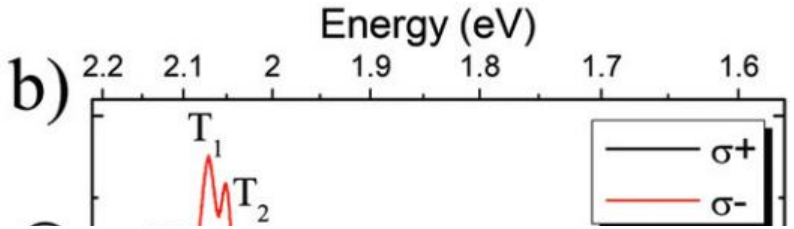
Under a critical SV concentration, nitrogen compensation doping can reduce the electron effective mass, which provides an effective way to improve the carrier mobility of WS_2

Summary of the FETs in various reports

Device structures	Threshold voltage [V]	Mobility [$cm^2 V^{-1} s^{-1}$]	On/Off ratio	Contact barrier	Ref.
WS_2-Al_2O_3/SiO_2	–	83	–	–	[19]
WS_2-Al_2O_3	–	–	$\approx 10^5$	Exist	[45]
WS_2-SiO_2	≈10	≈34.7	$\approx 10^6$	Exist	[46]
WS_2-SiO_2	–	<10	$\approx 10^5$	Exist	[47]
WS_2-SiO_2	–60–5	125	$\approx 10^5$–10^6	Exist	[48]
WS_2-SiO_2	≈30	50.5	10^7	–	[49]
WS_2-SiO_2	–	41.63	$\approx 10^3$	–	[50]
WS_2-SiO_2	>10	7.5	$\approx 10^5$	–	[51]
Graphene-WS_2-SiO_2	–	≈50	$\approx 10^7$	–	[52]
WS_2-SiO_2	≈30	≈50	$\approx 10^6$	Exist	[23]
WS_2-SiO_2	–	23.4	$\approx 10^7$	Exist	[53]
PMMA-WS_2-Al_2O_3/SiO_2, with nitrogen patching	–3.8	184.2	$\approx 10^5$–10^6	Disappear	This work

3

Revealing the nature of low-temperature photoluminescence peaks by laser treatment in van der Waals epitaxially grown WS_2 monolayers†



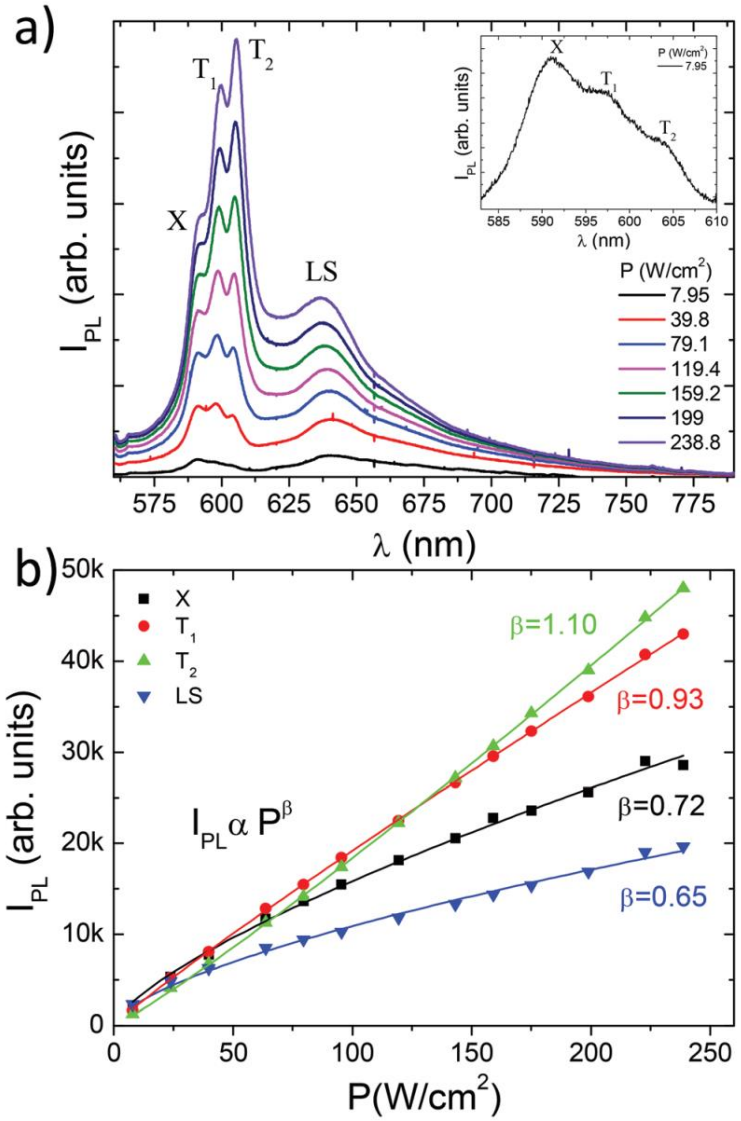
591nm/2.10eV X (neutral exciton state)

V. Orsi Gordo, [ab] M. A. G. Balanta, [ac] Y. Galvão Gobato, [*a] F. S. Covre, [a] H. V. A. Galeti, [d] F. Iikawa, [b] O. D. D. Couto, Jr., [b] F. Qu, [e] M. Henini, [fg] D. W. Hewak [h] and C. C. Huang [*h]

Strong emission bands on the lower energy side of the trion emission is because radiative recombination of biexcitons.

https://doi.org/10.1039/C8NR00719E

Macro-PL : 532nm excite, T=13K, σ_- excitation, spot 400 µm

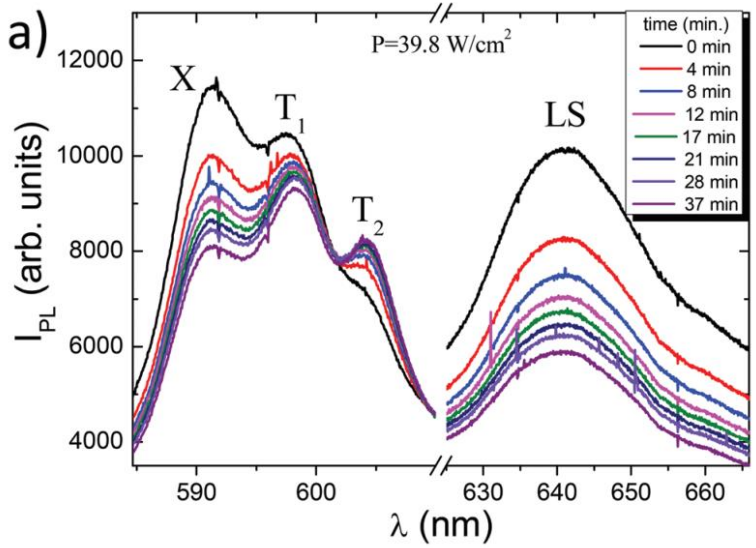


The rise of T_2 to a dark trion brightening process?

Energy separation between the T_1 and T_2 emissions expected for the dark trion process is $2\Delta_{so}$.

However, given the experimental value of 20 meV, Δ_{so} value is considerably smaller than the predicted one.

PL with time

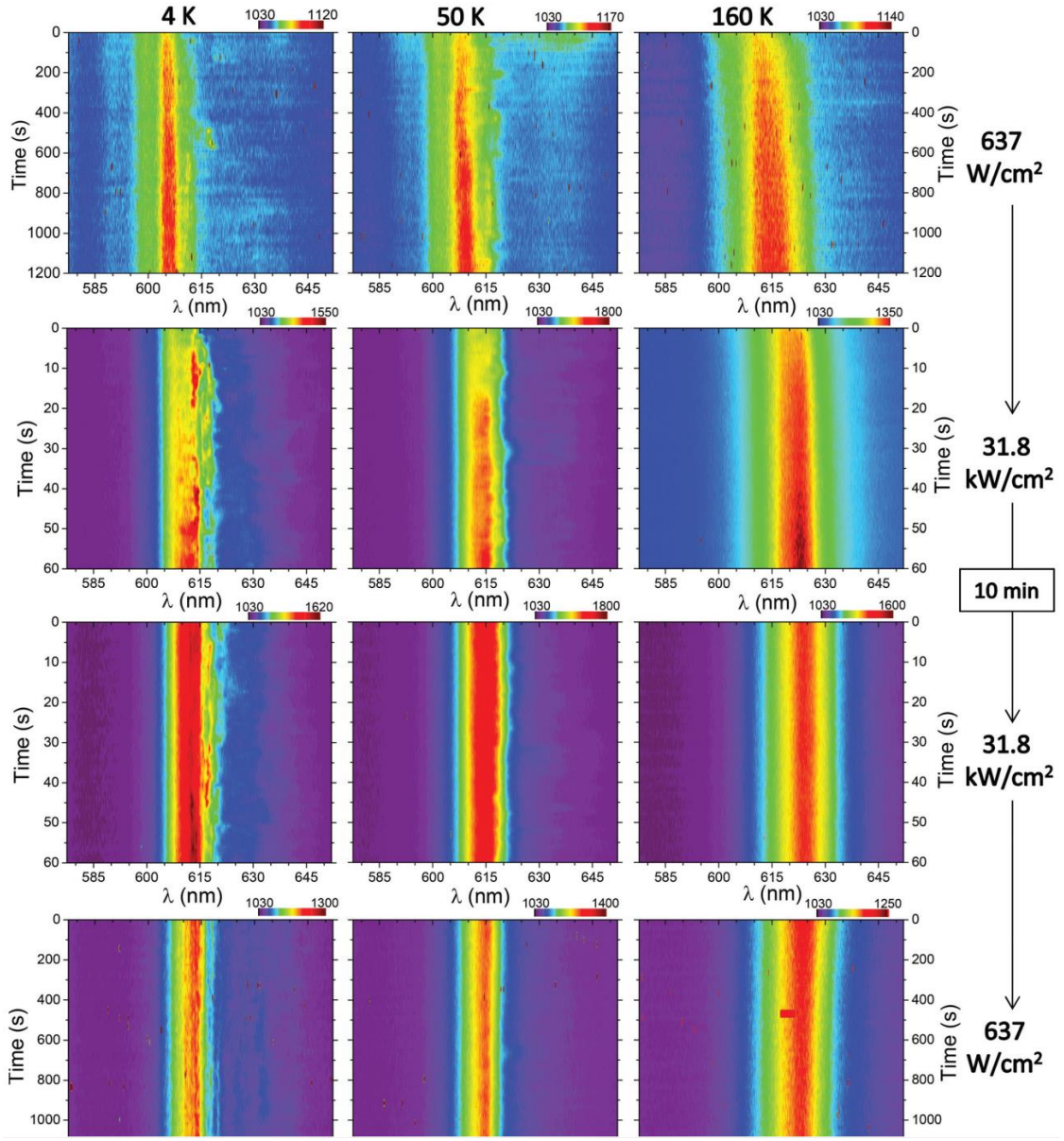


The increase of T_2 emission with time to an enhanced probability that electrons occupy the upper conduction sub band states due to high electron doping induced by laser irradiation.

T_1 and T_2 emissions increase at very similar rates when electrons are added to the system. This behavior further corroborates our attribution of T_2 to a trion state rather than a XX one.

As also seen in (b), X emission is more independent of the changes in LS. The absolute decrease in X intensity is attributed to the formation of charged complexes as the system is further doped with increase of time.

μ-PL spot 2 μm, in vacuum with He cryostat



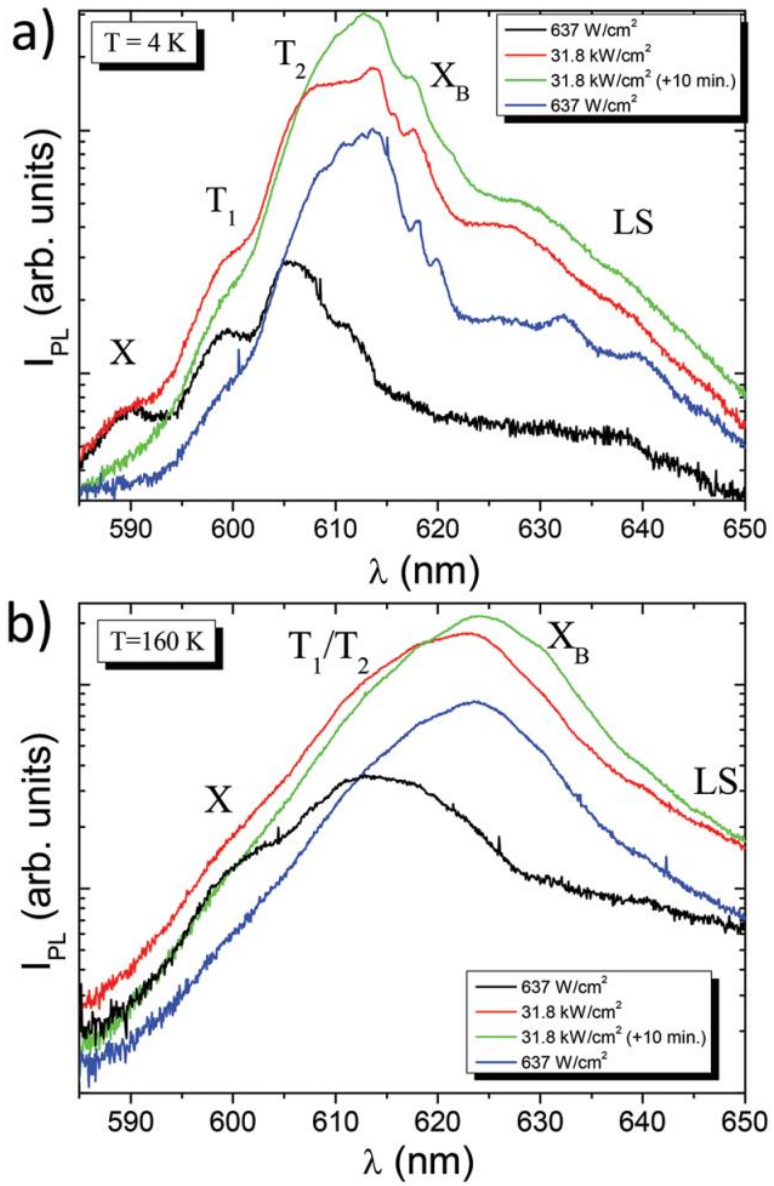
1. sample is subjected to a laser, then the PL emission is tracked for 20 minutes.
2. power increased and the measurement is performed once more (for 60 seconds).
3. laser incidence is maintained on the sample and after 10 minutes, measurement is repeated.
4. power is decreased and the PL is measured once more during 20 minutes.

Power up---redshift

After 10 mins, slight red shift and signal intensity increase

Power down---no blue shift and intensity bigger than first

Time integrated μ-PL spectra



(a)Neutral exciton emission is quenched (blue curve), while the spectrum is dominated by T_2 and X_B emissions after laser doping.

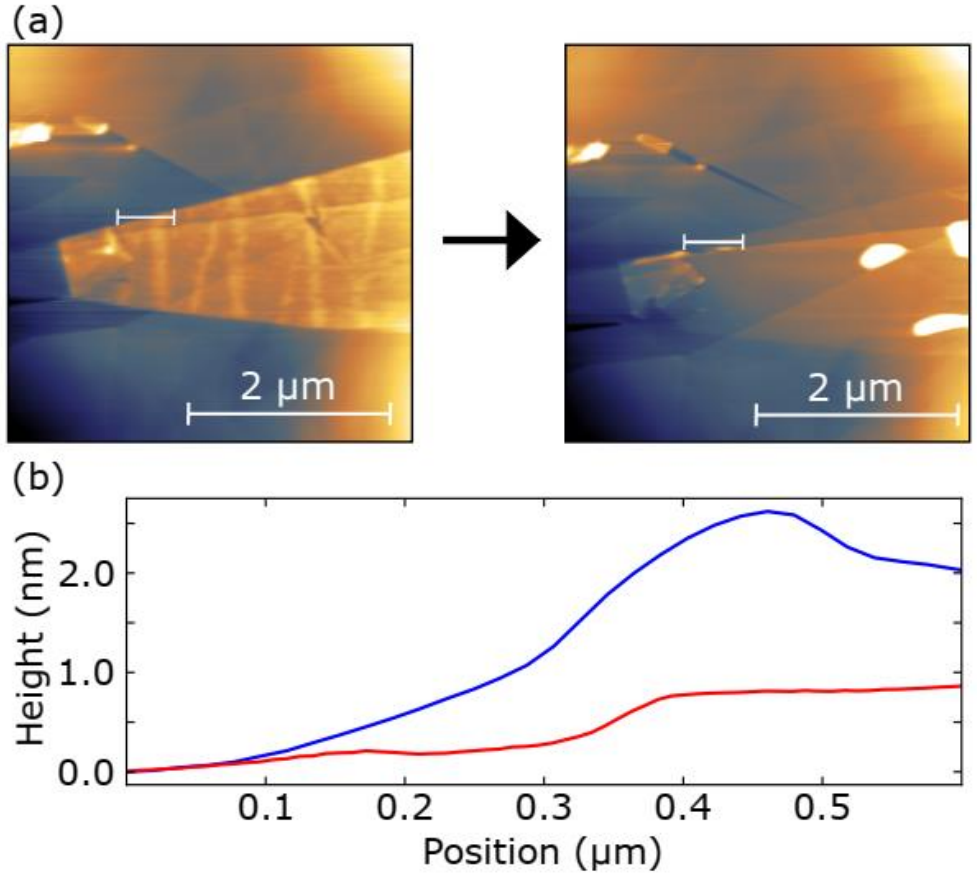
(b)160K spectrum is broaden and redshift, and after high power laser, the X_B peak will disappear.

And after recover to low power, the spectrum will blueshift (dominated by a broad emission which superimposes trions trions and X_B emission bands).

4

Single atom chemical identification of TMD defects in ambient conditions

Dunn E. J.[1,2], Young R. J.[1] and Jarvis S. P.[1]



Fabrication and Flattening

Sample preparation : mechanically exfoliated onto Nitto Blue Tapes (Acrylic/PVC) where repeated exfoliation dispersed and thinned the flakes.

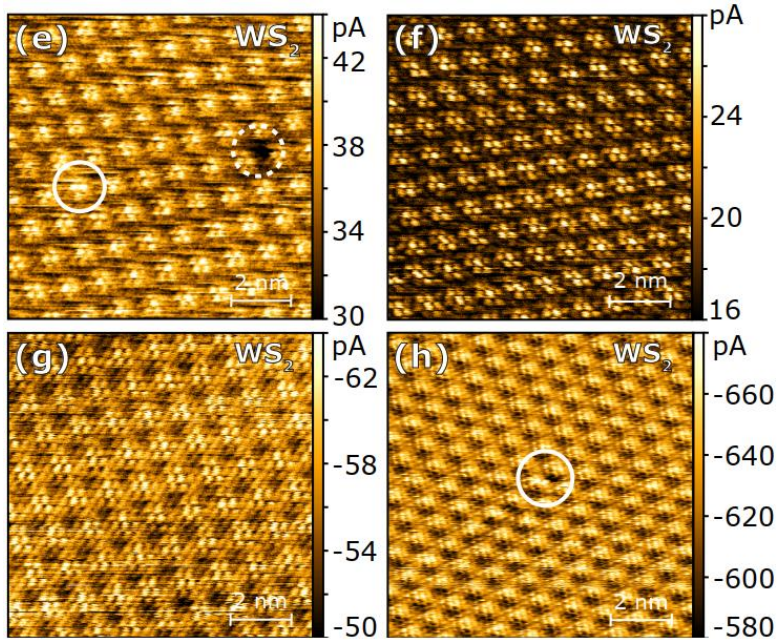
The flakes were then exfoliated from the tapes onto pre-prepared PDMS stamps affixed to glass slides where flakes were then assessed for quality using an optical microscope.

Plasma treatment : Samples were prepared in a glass barrelled plasma reactor, which was a QVF process pipe – dimensions 500mm length, 100 mm diameter, clamped between two steel end plates which serve as ground electrodes.

Plasma was ignited using a RFGC-100-13 power generator and automatic matching network , operated at a frequency of 13.56 MHz, connected to a 1 cm thick copper braid wrapped around the glass barrel three times.

https://arxiv.org/abs/2406.18324

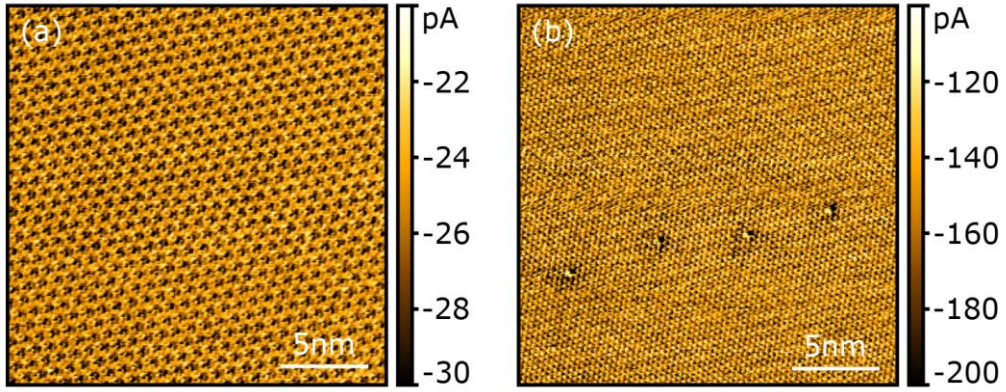
C(conductive)-AFM measurements of defects



(e) Augmenting (white circle) depleting defects(dashed white circle)

(f) 0.5 V and **(g)** -1 V sample bias in another region show only Moiré pattern

(h) This defect appears to change lattice site during scanning (top-to-bottom), showing a ‘slice’ effect where the defect switches from a dark to a bright feature. That may be a surface adsorbate. (hard to identify such defects against the background of the Moiré pattern)



Initially, at bias voltage of -0.4 V (a), the defects cannot be seen, then when the bias voltage is increased to -0.8 V (b) they can be made out.